

DATA ENGINEERING – ASSIGNMENT 1

Apache Hadoop MapReduce

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Topic	Apache Hadoop MapReduce

1. Aim

To configure and run Apache Hadoop in Ubuntu and use Hadoop MapReduce to process web-server access log data, calculate the total bandwidth used by each IP address, and display the MapReduce output.

2. Problem Statement

Large web-server log files contain many records showing IP addresses, requested URLs, status codes, and response sizes. Manually calculating the total data transferred by each IP address is time-consuming. The problem is to store the access log in HDFS and use a MapReduce program to efficiently group the records by IP address and calculate the total bandwidth used by each IP.

3. Work Performed in Ubuntu

The screenshots in the submitted PPT show the following practical work:

Hadoop configuration: The Hadoop configuration files `core-site.xml`, `hdfs-site.xml`, `mapred-site.xml` and `yarn-site.xml` were opened/edited. SSH was enabled and an RSA key was generated for Hadoop operation.

Starting Hadoop services: The Hadoop services were started using `start-dfs.sh` and `start-yarn.sh`. The `jps` command was used to check Hadoop processes such as NameNode, DataNode, SecondaryNameNode, ResourceManager, NodeManager and JobHistoryServer.

Preparing the input data: An `access.log` file was created. A Hadoop directory `/user/$USER/web_logs` was created in HDFS and `access.log` was uploaded using `hdfs dfs -put`. The stored file was checked using `hdfs dfs -cat`.

Running MapReduce locally: The `mrjob` package was installed and `bandwidth_read.py` was used with `access.log`. The program produced total bandwidth values for each IP address.

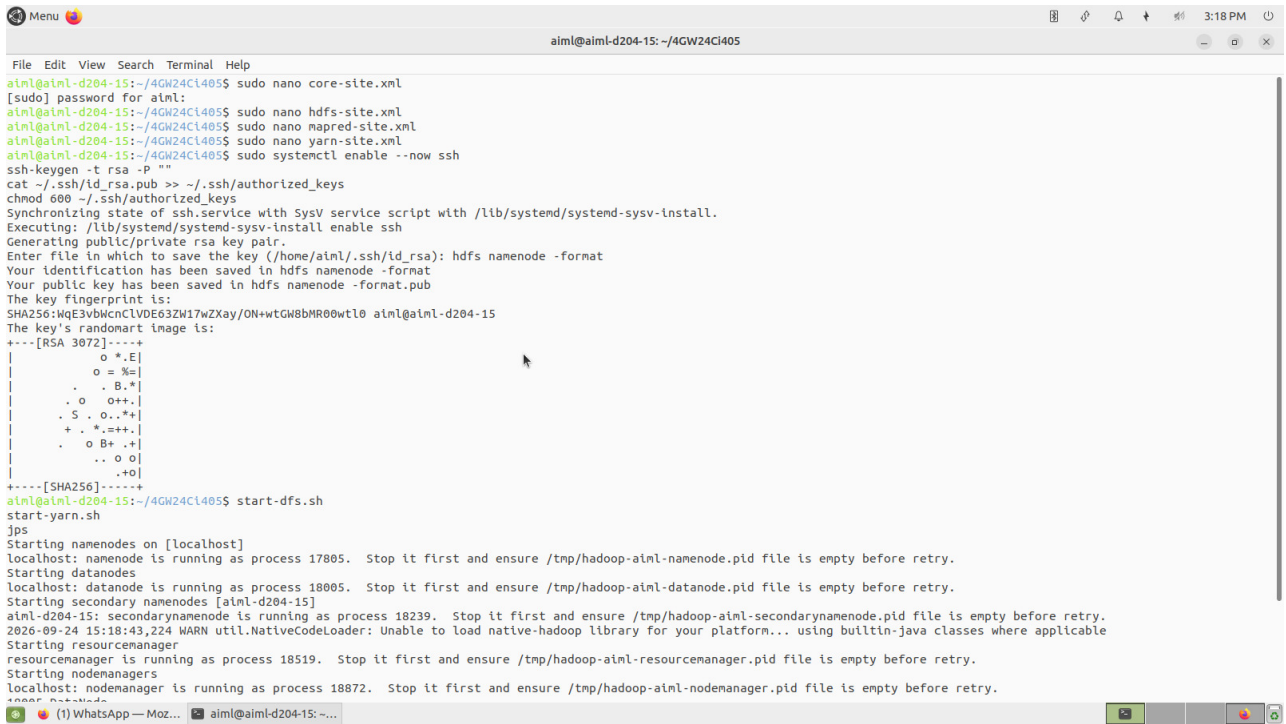
Running MapReduce on Hadoop: The program was executed using the Hadoop runner with the HDFS input path. Hadoop Streaming 3.3.6 was detected, the job was submitted through YARN, and the MapReduce job completed successfully.

Final result: The output showed total bandwidth values for five IP addresses: 192.168.1.10 – 21840, 192.168.1.20 – 27800, 192.168.1.30 – 46800, 192.168.1.40 – 11700, and 192.168.1.50 – 14060.

4. Screenshot-wise Explanation

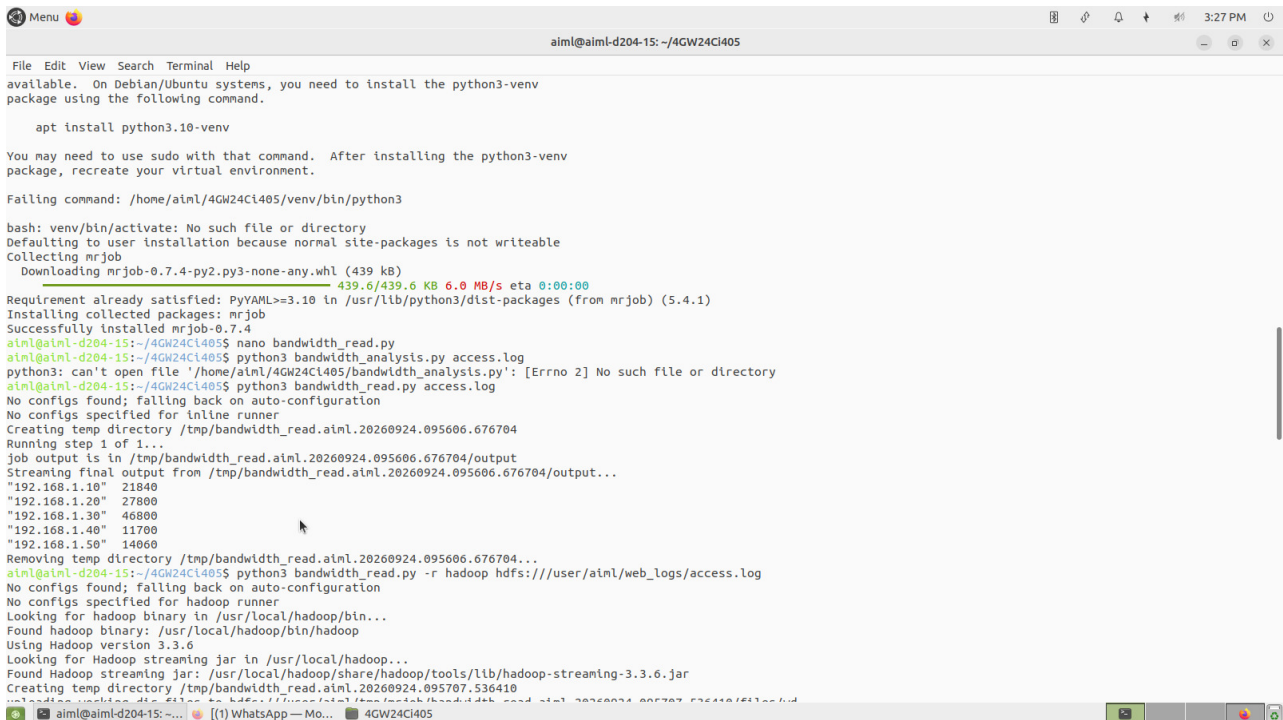
Screenshot 1 – Hadoop setup and services

The screenshot shows editing of the Hadoop XML configuration files, enabling SSH, generating an RSA key, formatting/initializing the Hadoop environment, and starting HDFS and YARN. The `jps`-related output confirms that the main Hadoop services were already running.



Screenshot 2 – Python/mrjob setup and local execution

The screenshot shows installation of mrjob and creation of bandwidth_read.py. An initial command using bandwidth_analysis.py failed because that file was not found. The correct bandwidth_read.py program was then executed on access.log and produced bandwidth totals for the IP addresses.



```
Menu
aiml@aiml-d204-15: ~/4GW24Ci405

File Edit View Search Terminal Help
available. On Debian/Ubuntu systems, you need to install the python3-venv
package using the following command.

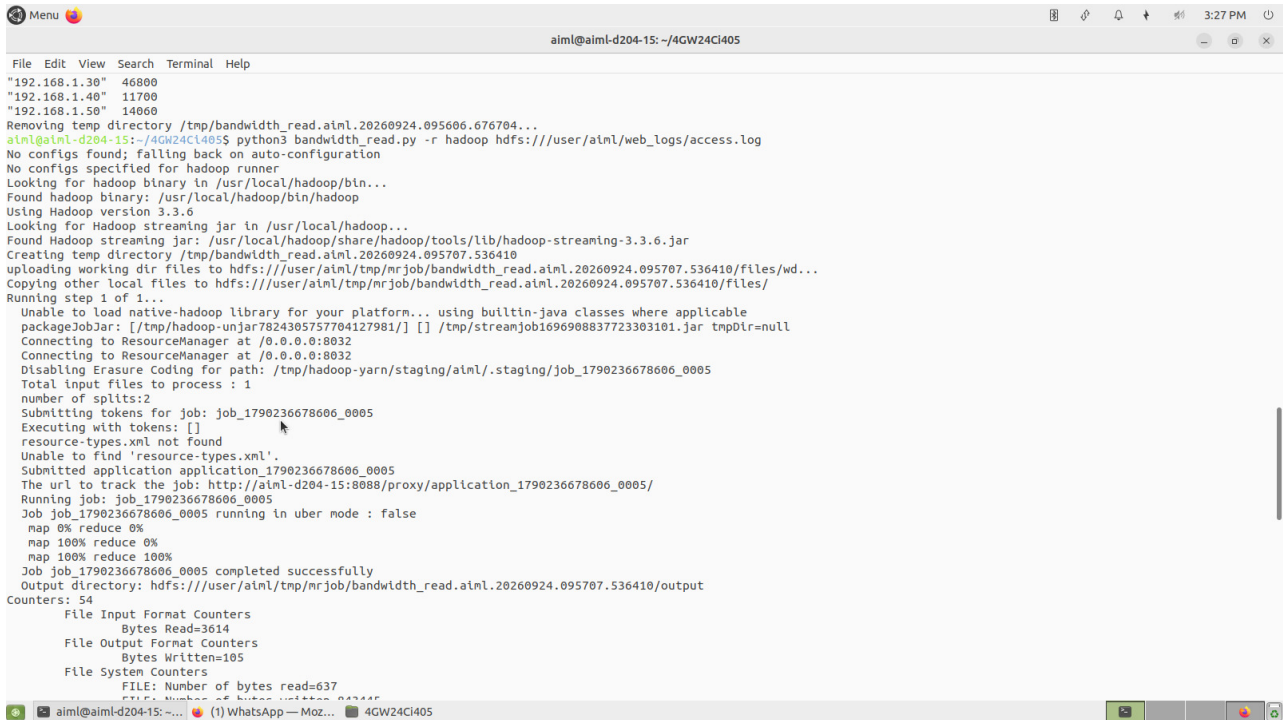
apt install python3.10-venv

You may need to use sudo with that command. After installing the python3-venv
package, recreate your virtual environment.

Falling command: /home/aiml/4GW24Ci405/venv/bin/python3
bash: venv/bin/activate: No such file or directory
Defaulting to user installation because normal site-packages is not writeable
Collecting mrjob
  Downloading mrjob-0.7.4-py2.py3-none-any.whl (439 kB)
    439.6/439.6 KB 6.0 MB/s eta 0:00:00
Requirement already satisfied: PyYAML>=3.10 in /usr/lib/python3/dist-packages (from mrjob) (5.4.1)
Installing collected packages: mrjob
Successfully installed mrjob-0.7.4
aiml@aiml-d204-15:~/4GW24Ci405$ nano bandwidth_read.py
aiml@aiml-d204-15:~/4GW24Ci405$ python3 bandwidth_analysis.py access.log
python3: can't open file '/home/aiml/4GW24Ci405/bandwidth_analysis.py': [Errno 2] No such file or directory
aiml@aiml-d204-15:~/4GW24Ci405$ python3 bandwidth_read.py access.log
No configs found; falling back on auto-configuration
No configs specified for inline runner
Creating temp directory /tmp/bandwidth_read.aiml.20260924.095606.676704
Running step 1 of 1...
Job output is in /tmp/bandwidth_read.aiml.20260924.095606.676704/output
Streaming final output from /tmp/bandwidth_read.aiml.20260924.095606.676704/output...
"192.168.1.10" 21840
"192.168.1.20" 27800
"192.168.1.30" 46800
"192.168.1.40" 11700
"192.168.1.50" 14060
Removing temp directory /tmp/bandwidth_read.aiml.20260924.095606.676704...
aiml@aiml-d204-15:~/4GW24Ci405$ python3 bandwidth_read.py -r hadoop hdfs:///user/aiml/web_logs/access.log
No configs found; falling back on auto-configuration
No configs specified for hadoop runner
Looking for hadoop binary in /usr/local/hadoop/bin...
Found hadoop binary: /usr/local/hadoop/bin/hadoop
Using Hadoop version 3.3.6
Looking for Hadoop streaming jar in /usr/local/hadoop...
Found Hadoop streaming jar: /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar
Creating temp directory /tmp/bandwidth_read.aiml.20260924.095707.536410
aiml@aiml-d204-15:~/4GW24Ci405$
```

Screenshot 3 – Hadoop execution begins

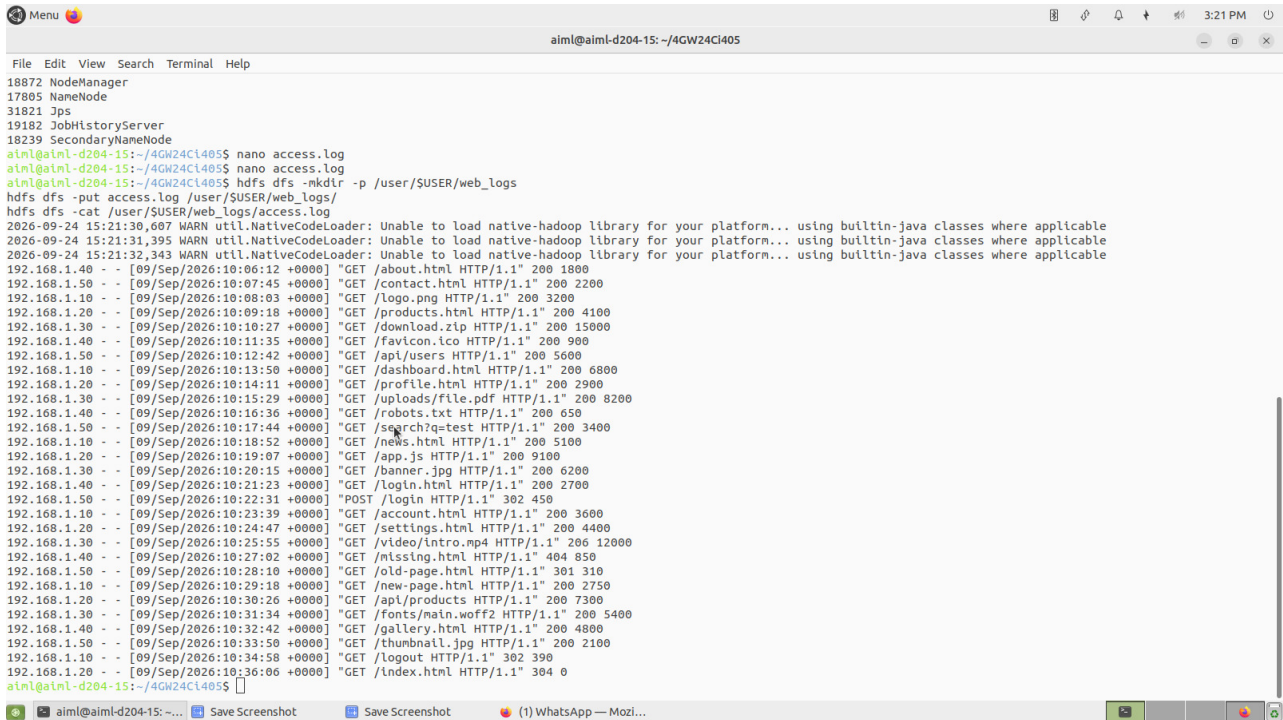
The Hadoop runner is used with the HDFS access-log path. Hadoop 3.3.6 and the Hadoop Streaming JAR are detected, the working files are uploaded to HDFS, and a MapReduce application is submitted to YARN.



```
Menu
aiml@aiml-d204-15: ~/4GW24C1405
File Edit View Search Terminal Help
"192.168.1.30" 46800
"192.168.1.40" 11700
"192.168.1.50" 14000
Removing temp directory /tmp/bandwidth_read.aiml.20260924.095606.676704...
aiml@aiml-d204-15:~/4GW24C1405$ python3 bandwidth_read.py -r hadoop hdfs:///user/aiml/web_logs/access.log
No configs found; falling back on auto-configuration
No configs specified for hadoop runner
Looking for hadoop binary in /usr/local/hadoop/bin...
Found hadoop binary: /usr/local/hadoop/bin/hadoop
Using Hadoop version 3.3.6
Looking for Hadoop streaming jar in /usr/local/hadoop...
Found Hadoop streaming jar: /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar
Creating temp directory /tmp/bandwidth_read.aiml.20260924.095707.536410
uploading working dir files to hdfs:///user/aiml/tmp/nrjob/bandwidth_read.aiml.20260924.095707.536410/files/wd...
Copying other local files to hdfs:///user/aiml/tmp/nrjob/bandwidth_read.aiml.20260924.095707.536410/files/
Running step 1 of 1...
Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
packageJobJar: [/tmp/hadoop-unjar7824305757704127981/] [] /tmp/streamjob1696908837723303101.jar tmpDir=null
Connecting to ResourceManager at /0.0.0.0:8032
Connecting to ResourceManager at /0.0.0.0:8032
Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/aiml/.staging/job_1790236678606_0005
Total input files to process : 1
number of splits:2
Submitting tokens for job: job_1790236678606_0005
Executing with tokens: []
resource-types.xml not found
Unable to find 'resource-types.xml'.
Submitted application application_1790236678606_0005
The url to track the job: http://aiml-d204-15:8088/proxy/application_1790236678606_0005/
Running job: job_1790236678606_0005
Job job_1790236678606_0005 running in uber mode : false
  map 0% reduce 0%
  map 100% reduce 0%
  map 100% reduce 100%
Job job_1790236678606_0005 completed successfully
Output directory: hdfs:///user/aiml/tmp/nrjob/bandwidth_read.aiml.20260924.095707.536410/output
Counters: 54
  File Input Format Counters
    Bytes Read=3614
  File Output Format Counters
    Bytes Written=105
  File System Counters
    FILE: Number of bytes read=637
    FILE: Number of bytes written=823445
```

Screenshot 4 – MapReduce job completion

The screenshot shows the MapReduce job progressing from map 0%/reduce 0% to map 100%/reduce 100%. The job completed successfully and Hadoop reports its output directory and counters.

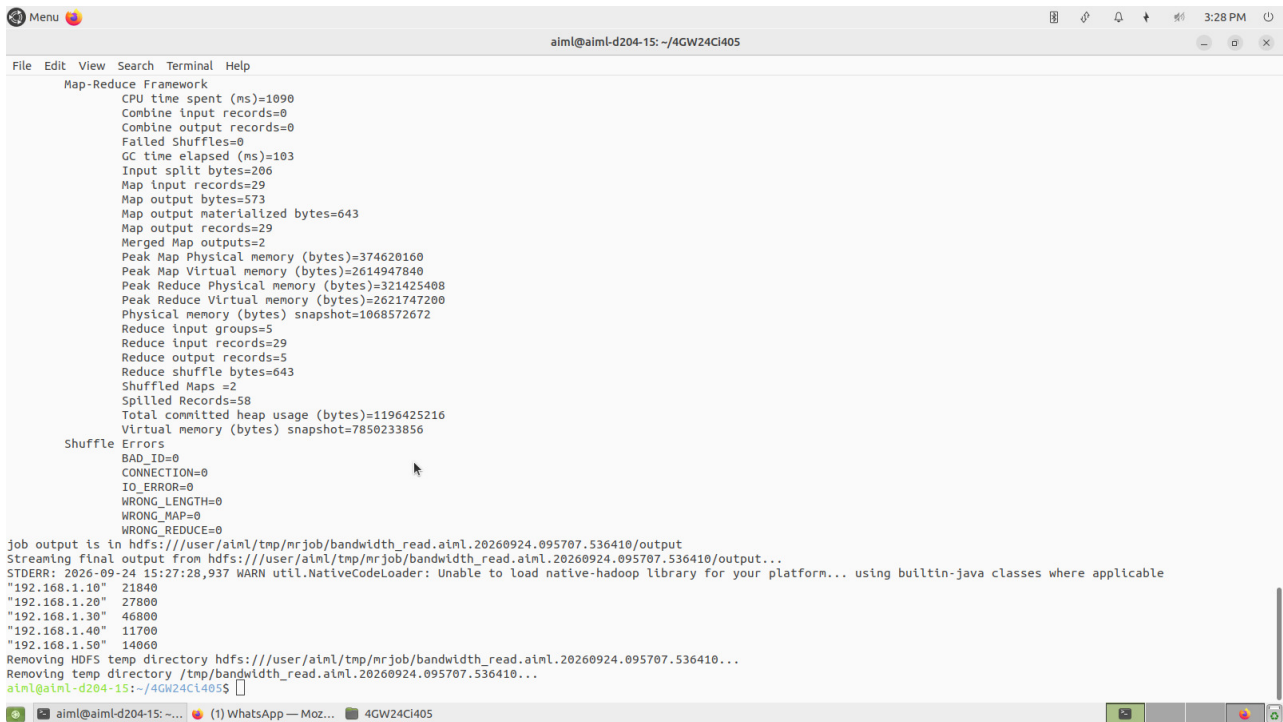


The screenshot shows a terminal window titled 'aiml@aiml-d204-15: ~/4GW24C1405'. The terminal displays the output of a Hadoop MapReduce job. The job has completed successfully, showing 'map 100%/reduce 100%'. The output directory is reported as '/user/\$USER/web_logs'. The terminal also shows various warnings and status messages, including 'WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable'. The bottom of the terminal shows the prompt 'aiml@aiml-d204-15:~/4GW24C1405\$'.

```
18872 NodeManager
17805 NameNode
31821 Jps
19182 JobHistoryServer
18239 SecondaryNameNode
aiml@aiml-d204-15:~/4GW24C1405$ nano access.log
aiml@aiml-d204-15:~/4GW24C1405$ nano access.log
aiml@aiml-d204-15:~/4GW24C1405$ hdfs dfs -mkdir -p /user/$USER/web_logs
hdfs dfs -put access.log /user/$USER/web_logs/
hdfs dfs -cat /user/$USER/web_logs/access.log
2026-09-24 15:21:30,607 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-09-24 15:21:31,395 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-09-24 15:21:32,343 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
192.168.1.40 - - [09/Sep/2026:10:06:12 +0000] "GET /about.html HTTP/1.1" 200 1800
192.168.1.50 - - [09/Sep/2026:10:07:45 +0000] "GET /contact.html HTTP/1.1" 200 2200
192.168.1.10 - - [09/Sep/2026:10:08:03 +0000] "GET /logo.png HTTP/1.1" 200 3200
192.168.1.20 - - [09/Sep/2026:10:09:18 +0000] "GET /products.html HTTP/1.1" 200 4100
192.168.1.30 - - [09/Sep/2026:10:10:27 +0000] "GET /download.zip HTTP/1.1" 200 15000
192.168.1.40 - - [09/Sep/2026:10:11:35 +0000] "GET /favicon.ico HTTP/1.1" 200 900
192.168.1.50 - - [09/Sep/2026:10:12:42 +0000] "GET /api/users HTTP/1.1" 200 5600
192.168.1.10 - - [09/Sep/2026:10:13:50 +0000] "GET /dashboard.html HTTP/1.1" 200 6800
192.168.1.20 - - [09/Sep/2026:10:14:11 +0000] "GET /profile.html HTTP/1.1" 200 2900
192.168.1.30 - - [09/Sep/2026:10:15:29 +0000] "GET /uploads/file.pdf HTTP/1.1" 200 8200
192.168.1.40 - - [09/Sep/2026:10:16:36 +0000] "GET /robots.txt HTTP/1.1" 200 650
192.168.1.50 - - [09/Sep/2026:10:17:44 +0000] "GET /search?q=test HTTP/1.1" 200 3400
192.168.1.10 - - [09/Sep/2026:10:18:52 +0000] "GET /news.html HTTP/1.1" 200 5100
192.168.1.20 - - [09/Sep/2026:10:19:07 +0000] "GET /app.js HTTP/1.1" 200 9100
192.168.1.30 - - [09/Sep/2026:10:20:15 +0000] "GET /banner.jpg HTTP/1.1" 200 6200
192.168.1.40 - - [09/Sep/2026:10:21:23 +0000] "GET /login.html HTTP/1.1" 200 2700
192.168.1.50 - - [09/Sep/2026:10:22:31 +0000] "POST /login HTTP/1.1" 302 450
192.168.1.10 - - [09/Sep/2026:10:23:39 +0000] "GET /account.html HTTP/1.1" 200 3600
192.168.1.20 - - [09/Sep/2026:10:24:47 +0000] "GET /settings.html HTTP/1.1" 200 4400
192.168.1.30 - - [09/Sep/2026:10:25:55 +0000] "GET /video/intro.mp4 HTTP/1.1" 206 12000
192.168.1.40 - - [09/Sep/2026:10:27:02 +0000] "GET /missing.html HTTP/1.1" 404 850
192.168.1.50 - - [09/Sep/2026:10:28:10 +0000] "GET /old-page.html HTTP/1.1" 301 310
192.168.1.10 - - [09/Sep/2026:10:29:18 +0000] "GET /new-page.html HTTP/1.1" 200 2750
192.168.1.20 - - [09/Sep/2026:10:30:26 +0000] "GET /api/products HTTP/1.1" 200 7300
192.168.1.30 - - [09/Sep/2026:10:31:34 +0000] "GET /fonts/main.woff2 HTTP/1.1" 200 5400
192.168.1.40 - - [09/Sep/2026:10:32:42 +0000] "GET /gallery.html HTTP/1.1" 200 4800
192.168.1.50 - - [09/Sep/2026:10:33:50 +0000] "GET /thumbnail.jpg HTTP/1.1" 200 2100
192.168.1.10 - - [09/Sep/2026:10:34:58 +0000] "GET /logout HTTP/1.1" 302 390
192.168.1.20 - - [09/Sep/2026:10:36:06 +0000] "GET /index.html HTTP/1.1" 304 0
aiml@aiml-d204-15:~/4GW24C1405$
```

Screenshot 5 – MapReduce output

The MapReduce framework counters show 29 map input records and 29 reduce input records, with 5 reduce output records. The final output contains the total bandwidth for five IP addresses.

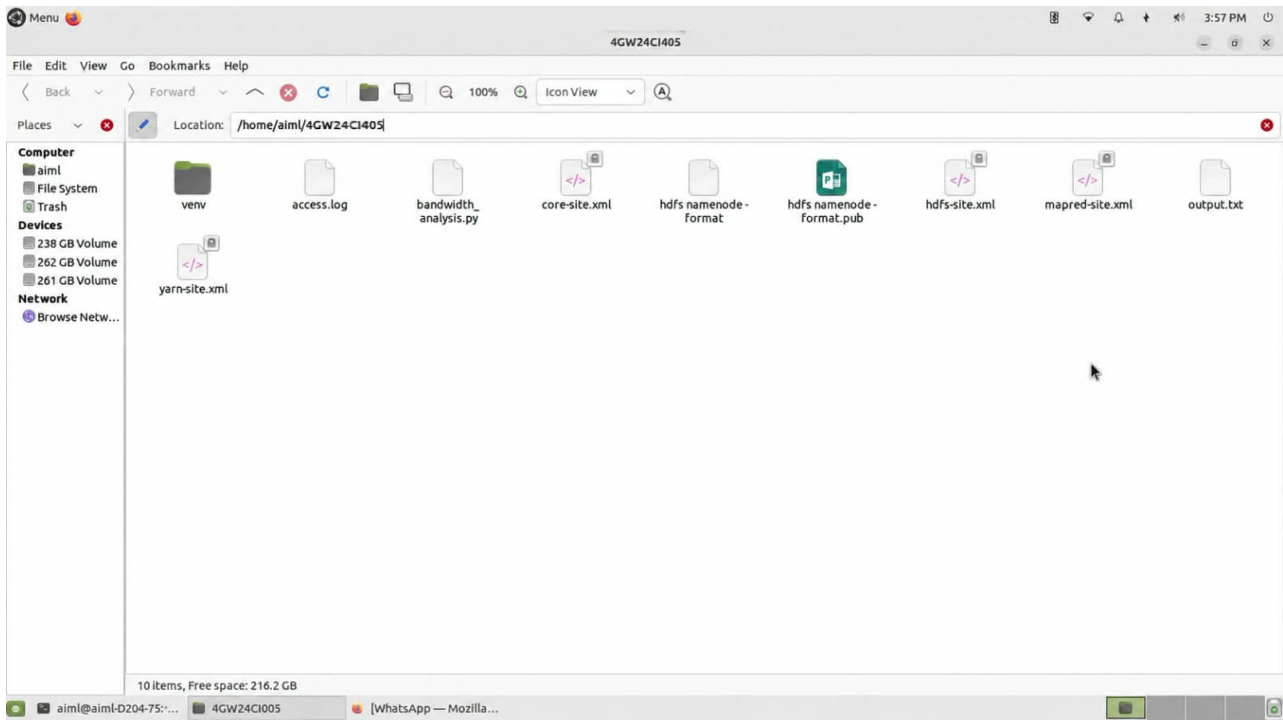


```
Menu
aiml@aiml-d204-15: ~/4GW24CI405

File Edit View Search Terminal Help
Map-Reduce Framework
  CPU time spent (ms)=1090
  Combine input records=0
  Combine output records=0
  Failed Shuffles=0
  GC time elapsed (ms)=103
  Input split bytes=206
  Map input records=29
  Map output bytes=573
  Map output materialized bytes=643
  Map output records=29
  Merged Map outputs=2
  Peak Map Physical memory (bytes)=374620160
  Peak Map Virtual memory (bytes)=2614947840
  Peak Reduce Physical memory (bytes)=321425408
  Peak Reduce Virtual memory (bytes)=2621747200
  Physical memory (bytes) snapshot=1068572672
  Reduce input groups=5
  Reduce input records=29
  Reduce output records=5
  Reduce shuffle bytes=643
  Shuffled Maps =2
  Spilled Records=58
  Total committed heap usage (bytes)=1196425216
  Virtual memory (bytes) snapshot=7850233856
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
job output is in hdfs:///user/aiml/tmp/mrjob/bandwidth_read.aiml.20260924.095707.536410/output
Streaming final output from hdfs:///user/aiml/tmp/mrjob/bandwidth_read.aiml.20260924.095707.536410/output...
STDERR: 2026-09-24 15:27:28,937 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
"192.168.1.10" 21840
"192.168.1.20" 27800
"192.168.1.30" 46800
"192.168.1.40" 11700
"192.168.1.50" 14060
Removing HDFS temp directory hdfs:///user/aiml/tmp/mrjob/bandwidth_read.aiml.20260924.095707.536410...
Removing temp directory /tmp/bandwidth_read.aiml.20260924.095707.536410...
aiml@aiml-d204-15:~/4GW24CI405$
```

Screenshot 6 – Project files

The Ubuntu file manager shows the project directory containing files such as access.log, bandwidth_read.py, core-site.xml, hdfs-site.xml, mapred-site.xml, output.txt and other Hadoop-related files.



5. Final Result

IP Address	Total Bandwidth
192.168.1.10	21840
192.168.1.20	27800
192.168.1.30	46800
192.168.1.40	11700
192.168.1.50	14060

The practical demonstrates the use of HDFS for storing the web log and Hadoop MapReduce for processing the log data. The Hadoop job completed successfully and generated five grouped results, one for each IP address present in the input records.

Note: The explanations above are based on the visible content of the submitted PPT screenshots. No extra implementation details have been assumed where the screenshots do not show them.